ACADEMIC INTELLIGENCE PLATFORM

EduVantage AI

Student Success Intelligence Platform for identifying academic risk, understanding learning behavior, and guiding timely human-led interventions.

OULAD Ingestion — **Temporal Signals** — **Agentic Audit**

Human Action

Executive & Technical Panel Briefing · Open University Learning Analytics Dataset (OULAD)



Made with GAMMA

STUDENT SUPPORT

Student support is buried in fragmented learning records

- Academic teams must connect assessment performance, learning activity, registration history, and previous outcomes to understand student progress.
- Manual review makes it difficult to prioritize who needs support first.
- Goal: move from retrospective searching to evidence-based, explainable intervention.



Made with GAMMA

CORE DATA ENGINE

One intelligence layer turns OULAD data into action

From 32,593 raw student course records to structured profiles, semantic vectors, and grounded reports.

01

1. Ingest Multi-Source Data

Cleans and unifies anonymized student demographic files, assessment submissions, registration timings, and daily VLE engagement clicks from OULAD.

OULAD Tables

02

2. Synthesize & Embed

Constructs rolling temporal features, velocity metrics, and semantic vector embeddings, creating searchable historical case baselines for similar student patterns.

pgvector Index

03

3. Surface Grounded Reports

Delivers a unified report with risk tiering, factual evidence logs, similar student outcomes, recommended actions, confidence scores, and known limitations.

Advisor Deliverable

Advisory principle: Every output pairs statistical indicators with explicit audit evidence and confidence bounds, preventing algorithmic overreach.

Made with GAMMA

SYSTEM TOPOLOGY

The platform separates experience, intelligence, and evidence layers

Experience

Next.js dashboard, student profiles, risk center, similarity search, reports, and AI assistant.

Application

FastAPI gateway with authentication, analytics, retrieval, risk, recommendation, agent, and evaluation services.

Evidence

PostgreSQL for structured records and audit logs, pgvector for embeddings, Redis for caching, and model services for prediction and explanation.

Governance

RBAC, input validation, privacy-aware embeddings, audit logging, monitoring, and human approval.

Designed as modular services so the demo can scale from deterministic DEMO_MODE to production providers.

Made with GAMMA

INTERVENTION PIPELINE

The architecture turns raw records into a grounded intervention

01

Ingest

Clean OULAD files and construct student, academic, engagement, registration, and outcome representations.

02

Retrieve

Combine keyword search and semantic embeddings, apply metadata filters, deduplicate, and rerank the top historical cases.

03

Reason

Performance, engagement, risk, similarity, and intervention agents analyze the evidence in sequence.

04

Validate

An LLM-as-Judge checks groundedness, completeness, safety, and recommendation quality before the educator sees the result.

Every answer exposes evidence, confidence, and limitations rather than presenting a prediction as certainty.

Made with GAMMA

DECISION SUPPORT & TRIAGE

Transparent risk signals make intervention prioritization practical

Interpretable features

Score trend, missed assessments, completion, activity frequency, engagement decline, previous attempts, credits, and registration history.

Risk prioritization

Classify students as low, moderate, high, or critical risk and rank an intervention queue.

Explain the “why”

Explain the “why” in educator language, for example declining scores, reduced activity, missed assessments, or recent inactivity.

Recommendations remain advisory

An advisor reviews, accepts, rejects, and records the outcome.

Made with GAMMA

VERIFICATION & METRICS

Demo flow and evaluation close the loop

DeepEval and LLM-as-Judge

End-to-End Advisor Demo Flow

1. Open the dashboard and identify high-risk or deteriorating students.
2. Inspect one student’s performance, engagement, risk explanation, and similar historical cases.
3. Generate an evidence-backed intervention recommendation and review its confidence and limitations.
4. Measure faithfulness, relevance, contextual precision, recall, hallucination rate, recommendation quality, and model performance with DeepEval and LLM-as-Judge.

Outcome: earlier, more consistent support decisions without automating high-impact decisions about students.

Evaluation Criteria

Faithfulness, relevance, contextual precision, recall, hallucination rate, recommendation quality, and model performance.

Measured with DeepEval and LLM-as-Judge.

Made with GAMMA